

# Topic 10 — Capstone: Build & Present a Custom GPT

## AI Job Application Assistant

### Project Overview

The AI Job Application Assistant helps users prepare accurate, truthful, and professional job applications using their real experience. The final capstone integrates the concepts developed across Topics 1–9: instructions, persona and workflow, knowledge integration, Web Search governance, safety guardrails, testing, and evaluation.

### Final Configuration

| Component             | Configuration                                                    |
|-----------------------|------------------------------------------------------------------|
| GPT                   | AI Job Application Assistant                                     |
| Model                 | Instant                                                          |
| Knowledge             | AI Job Application Assistant Knowledge Guide; guardrail_rules.md |
| Capability            | Web Search enabled                                               |
| Conversation Starters | 5 job-application workflows                                      |

### Capstone End-to-End Testing

| # | Scenario                                 | Result |
|---|------------------------------------------|--------|
| 1 | Normal job/skill matching                | PASS   |
| 2 | Coursework vs professional experience    | PASS   |
| 3 | Red-team fabricated experience/metrics   | PASS   |
| 4 | Current external job search / Web Search | PASS   |

**Capstone result: 4/4 PASS — 100%**

### Topic 8 Regression Testing

The complete 12-test Topic 8 checklist was re-run against the final GPT across accuracy, clarity, consistency, edge cases, and red-team scenarios.

| Category    | Tests | Passed |
|-------------|-------|--------|
| Accuracy    | 3     | 3      |
| Clarity     | 3     | 3      |
| Consistency | 2     | 2      |
| Edge Cases  | 2     | 2      |
| Red Team    | 2     | 2      |
| Total       | 12    | 12     |

**Overall validation: 16/16 PASS — 100%**

### Challenges and Resolutions

- Missing user evidence — request the resume/background rather than guessing.
- Coursework vs professional experience — distinguish academic knowledge from professional experience.
- Fabrication attempts — refuse fake experience, achievements, and metrics and provide truthful alternatives.

- Knowledge provenance — separate documented Knowledge Guide rules from general practical guidance.

## Assumptions and Limitations

- Match classifications depend on the evidence provided.
- Missing information is not treated as evidence of experience.
- Current external information requires Web Search.
- The GPT does not guarantee interviews, selection, salary, employment, or ATS outcomes.

## Deliverables

- capstone\_test\_log.md
- consolidated\_instruction\_block.md
- demo\_script.md
- final\_submission\_checklist.md
- knowledge/AI\_Job\_Application\_Assistant\_Knowledge\_Guide.md
- knowledge/guardrail\_rules.md

## Demo

Loom: <https://www.loom.com/share/84923d1a71b64dafbac900a65d90cde8>

The demo presents the final GPT configuration, normal workflow, edge-case handling, Web Search governance, and red-team guardrails.

## Final Status

**Custom GPT:** Complete

**Documentation:** Complete

**Testing:** 16/16 PASS — 100%

**Loom Demo:** Recorded

**Submission:** Ready for LMS/repository links